FACULDADE DE DESIGN,
TECNOLOGIA E COMUNICAÇÃO
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MASTER'S IN CREATIVE COMPUTING
AND ARTIFICIAL INTELLIGENCE

EEG-Based Remote Car Control Using the Emotiv EPOC X:

System Development and Experimental Evaluation

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Dissertação apresentada ao IADE - Faculdade de Design, Tecnologia e Comunicação da Universidade Europeia, para cumprimento dos requisitos necessários à obtenção do grau de Mestre em Creative Computing and Artificial Intelligence realizada sob a orientação científica da Doutora Pamela Teubig, professora assistente do *IADE* e de José Graça, professor do IADE.

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Keywords *[Keywords to add.]*

abstract *[Text to write.]*

Glossary

Term *[Definition to write.]*

List of Abbreviations

ABBR *[Meaning to write.]*

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1 Introduction

[Aim: open the dissertation in one or two short paragraphs. Cover: the topic of EEG-based control of a physical device; the Emotiv EPOC X and the remote car as the study platform; the open-source Python framework as the main output.]

1.1 Background and Context

[Aim: give the reader what they need to follow the problem. Cover: what brain-computer interfaces are; non-invasive EEG and the arrival of consumer headsets; vehicle and robot control as an application. Keep this brief and leave the detail to Chapter 2.]

1.2 Motivation

[Aim: explain why the work is worth doing now. Cover: hands-free and assistive control as a real need; lower cost of consumer EEG hardware; the lack of open, reproducible tooling; your own academic interest in the topic.]

1.3 Problem Statement

[Aim: define the problem precisely. Cover: mental commands from consumer EEG are noisy and vary between users; controlling a physical car needs commands that are dependable, timely and safe; it is unclear how far the EPOC X meets those demands. End with a single sentence stating the problem.]

1.4 Research Gap

[Aim: name the gap in brief. Cover: what existing work has shown; what remains missing, such as open frameworks or evaluations reporting reliability and latency together on a physical vehicle. Confirm each claim against the literature; the full evidence belongs in Section 2.13.]

1.5 Research Question

[Aim: state one main question that the experiment can answer. Starting point: to what extent can mental commands from the Emotiv EPOC X, processed by an open-source Python framework, reliably control a remote car in real time? Add sub-questions only if each maps to a metric in Section 4.11.]

1.6 Main Research Objective

[Aim: state the overall objective in one sentence. Starting point: to design, build and experimentally evaluate a brain-controlled car system, driven by Emotiv EPOC X mental commands through an open-source Python framework.]

1.7 Specific Objectives

[Aim: break the main objective into a numbered list. Cover: review the literature; build the car and its electronics; develop the Python framework and integrate it with the car; add safety mechanisms; define the protocol and metrics; run the experiments; analyse the results; publish the code. Start each with a verb, keep each measurable and note the chapter that delivers it, because Section 7.4 reports back on this list.]

1.8 Research Hypotheses

[Include only if applicable. Aim: state testable hypotheses tied to the metrics in Section 4.11. Possible forms: H1, command success rate reaches a stated threshold; H2, end-to-end response time stays within a stated limit; H3, performance changes across sessions. Justify each threshold from the literature or the pilot study, and give null forms if you use inferential statistics.]

1.9 Scope and Limitations

[Aim: set the boundaries of the study. Cover: what is included, such as one headset, Emotiv's built-in detection, one purpose-built car and a controlled environment; what is excluded, such as clinical users and real road use; known constraints such as sample size and time.]

1.10 Expected Contributions

[Aim: say what the work adds. Cover: the brain-controlled car system; the open-source framework; a reusable experimental protocol; a defined set of real-time performance metrics; evidence on the feasibility of consumer EEG control. Separate practical from academic contributions, and revisit them in Section 7.3.]

1.11 Dissertation Structure

[Aim: map the document. Cover: one sentence for each of Chapters 2 to 7 and the Appendices. Write this section last.]

2 Brain-Computer Interfaces and EEG-Based Control

[Aim: open the literature review in one short paragraph. Cover: the subject of the chapter and how it supports the research question.]

2.1 Introduction

[Aim: explain how the review is organised. Cover: the search approach, including databases, keywords and date range; what you included and excluded; the order of the sections that follow.]

2.2 Fundamentals of Brain-Computer Interfaces

[Aim: define the field. Cover: what a BCI is; the standard pipeline of acquisition, processing, translation and output; invasive versus non-invasive; active, reactive and passive BCIs; main application areas. Cite foundational sources.]

2.3 EEG Technology and Brain Signal Acquisition

[Aim: explain how EEG works and where it falls short. Cover: what EEG measures; frequency bands; electrode placement; wet versus dry or saline sensors; artefacts from muscle and eye movement; low spatial resolution and signal-to-noise problems.]

2.4 EEG-Based BCI Paradigms

[Aim: compare the main ways to generate commands. Cover: motor imagery; P300; SSVEP; other mental tasks; facial signals as non-brain alternatives. Compare training needs, command rate and suitability for continuous control, ideally in a table, and say which paradigm this work uses and why.]

2.5 Emotiv EPOC X: Architecture and Capabilities

[Aim: describe the device used in the study. Cover: channels and sensor layout, sampling rate, resolution, connectivity, sensors and software. Take every specification from Emotiv's documentation and cite it; also report independent validation studies against research-grade EEG.]

2.6 Emotiv Mental Commands: Principles and Limitations

[Aim: explain the detection feature your framework depends on. Cover: how users train commands; the neutral state; how many commands can be active; the output of command and power values; what is and is not public about the algorithm. Note the consequences of treating it as a black box.]

2.7 Consumer-Grade EEG Devices in BCI Applications

[Aim: place the EPOC X among similar devices. Cover: other consumer headsets; cost and usability benefits; accuracy limits compared with research systems; reported use in BCI studies.]

2.8 Machine Learning Approaches for EEG Classification

[Aim: review how EEG is classified. Cover: feature extraction; classical models such as LDA and SVM; deep learning such as CNN-based models; subject-dependent versus subject-independent training; small dataset and overfitting problems. Link this to the optional work in Section 4.13.]

2.9 EEG-Based Control of Vehicles and Mobile Robots

[Aim: review the closest applied work. Cover: studies of EEG-controlled cars, wheelchairs and robots; for each, the device, paradigm, number of commands, participants and results. A comparison table works well here and supports Section 6.8.]

2.10 Challenges of Real-Time BCI Control

[Aim: introduce the challenges that shape the framework and metrics. Cover: why online control is harder than offline classification, and how each subsection below links to a metric in Section 4.11.]

2.10.1 Command Detection and Classification Reliability

[Aim: review how reliable command detection is in practice. Cover: reported accuracy and success rates; variation between users and sessions; differences between offline and online figures; how studies measure reliability.]

2.10.2 Response Latency

[Aim: explain the delays in a BCI loop. Cover: the sources of delay, including signal windowing, processing, transmission and actuation; reported latencies; what counts as acceptable for vehicle control.]

2.10.3 False Activations and Missed Commands

[Aim: explain the two main error types. Cover: the difference between false activations and missed commands; why false activations matter most for a moving car; thresholding and the trade-off between the two errors.]

2.10.4 Cognitive Workload and User Adaptation

[Aim: cover the human side of control. Cover: fatigue and mental effort; BCI inefficiency, where some users cannot gain control; learning across sessions; how workload is measured, such as NASA-TLX.]

2.11 Safety and Reliability in BCI-Driven Systems

[Aim: review safety in brain-controlled devices. Cover: why noisy input calls for protection; emergency stops and fail-safe states; shared control; watchdogs and timeouts; ethical duties to users. Use this to justify the design in Section 3.11.]

2.12 Existing Frameworks and Related Work

[Aim: review available software. Cover: open-source BCI tools and Emotiv-based projects; what they support; their licence, maintenance and limitations. State what an existing tool would not give this project, justifying a new framework.]

2.13 Research Gap Summary

[Aim: close the chapter with the evidence for the gap. Cover: the main findings of the review; the specific gap this study addresses, more fully than Section 1.4; how the findings shape the framework design and experimental method.]

3 Development of an Emotiv EPOC X-Based Brain-Controlled Car System

[Aim: open the chapter in one short paragraph. Cover: the purpose of the brain-controlled car system and how this chapter answers the design objectives from Section 1.7.]

3.1 Introduction

[Aim: outline the chapter. Cover: the design goals, namely real-time control, safety, data capture and openness; the order of the sections that follow.]

3.2 System Requirements

[Aim: list what the system must do. Cover: functional requirements, such as acquiring commands, mapping them, driving the car, stopping and logging; non-functional requirements, such as latency, reliability and portability. Give each an identifier, such as FR1 or NFR1, so Section 3.13 can test it.]

3.3 Overall Architecture

[Aim: show the whole system at a glance. Cover: an architecture diagram from headset through the Python framework and microcontroller to the motors; the main modules and their responsibilities; data flow; why the system is modular. Place the diagram in figures/ch3.]

3.4 Emotiv EPOC X and Brain Activity Acquisition

[Aim: explain how brain activity enters the system. Cover: headset fitting and contact quality; authentication and app credentials; session creation; subscribing to the data stream; handling errors and disconnections. Follow the Cortex API documentation for the version you used, and never show real credentials.]

3.5 Car Construction and Integration of Electronic Components

[Aim: describe the design of the car. Cover: chassis, motors and steering; motor driver, power supply and wireless module; wiring, with a diagram and photos; a component table with part names and quantities; why each part was chosen. Section 4.3 records the car as used in the experiments, so keep the design reasoning here.]

3.6 Microcontroller, Firmware and Motor Control

[Aim: explain the embedded software. Cover: the microcontroller chosen and why; firmware structure and development tools; how commands become motor

signals, such as PWM speed control; start-up state and stopping behaviour. Use a short code listing if useful.]

3.7 Development of the Python Framework

[Aim: explain how the framework was built. Cover: Python version, libraries and module structure; concurrency, such as asyncio or threads; configuration; a table of software versions. Also cover its open-source design: repository structure, licence, dependency management, installation and how to extend it to other devices, with full details in Appendix E.]

3.8 Communication Between the Python Framework and the Microcontroller

[Aim: describe the link to the car. Cover: the wireless technology used; the message format; acknowledgements and retries; behaviour when the link drops. Explain why you chose this option over the alternatives.]

3.9 Data Processing and Identification of Commands

[Focus on Emotiv Mental Commands if that is the option you choose. Aim: explain what happens to a command on arrival. Cover: the received action and power values; timestamps; threshold filtering; smoothing or debouncing, such as requiring a command to persist. Justify each parameter and show a short code listing if useful.]

3.10 Mapping Commands to Car Movements

[Aim: explain how brain commands become car actions. Cover: the mapping table, such as push to forward; the role of neutral; handling of conflicting commands; state logic, ideally as a state diagram.]

3.11 Real-time Control and Safety Mechanisms

[Aim: explain how the system stays timely and safe. Cover: the control loop and update rate; buffering and avoiding stale commands; where timestamps are captured for latency measurement; a manual emergency stop independent of the EEG channel; stopping on lost signal or a timeout watchdog; default to stopped; speed limits. Show which risks each mechanism covers, drawing on Section 2.11.]

3.12 Data Logging and System Monitoring

[Aim: explain what the system records and displays. Cover: logged fields, such as timestamps, commands, power values and car actions; file format; session identifiers; how participant data stays pseudonymised; live monitoring of signal quality, connection and car state during sessions. Make sure the logs can produce every metric in Section 4.11.]

3.13 Technical Testing and Integration Validation

[Aim: show the system works before people use it. Cover: unit tests; bench tests of motors and firmware; integration tests from framework to car; a simulated command source for tests without a headset; timing and emergency stop tests; a table linking requirements from Section 3.2 to test results.]

3.14 Chapter Summary

[Aim: close the chapter. Cover: what was built, how it meets the requirements, and how it serves as the instrument for the evaluation in Chapter 4.]

4 Research Methodology and Experimental Evaluation

[Aim: open the chapter in one short paragraph. Cover: how the evaluation will answer the research question and test any hypotheses; the order of the sections that follow.]

4.1 Research Design

[Aim: name and justify the design. Cover: quantitative experimental evaluation, or a mixed approach if you add questionnaires; within-subject or repeated measures as applicable; independent and dependent variables; why this design fits the question.]

4.2 Experimental Setup and Equipment

[Aim: let another researcher rebuild the setup. Cover: headset and sensor preparation; the car; the computer and software versions; the physical arrangement, with a photo or diagram; the procedure for checking signal contact quality.]

4.3 Car Construction and Electronics Integration

[Aim: record the car as used in the experiments. Cover: the final configuration, with a photo; any changes made after Chapter 3; how consistent behaviour was checked before each session, such as battery level and motor response. Refer to Section 3.5 for the design rather than repeating it.]

4.4 Participants and Ethical Considerations

[Aim: describe who took part and how they were protected. Cover: recruitment; inclusion and exclusion criteria; number of participants and why; informed consent, with forms in Appendix B; ethics approval from the institution; data protection under GDPR; safety and the right to withdraw. If approval is pending, say so and do not state it as granted.]

4.5 Experimental Environment

[Aim: describe where the tests took place. Cover: the room or track; layout and dimensions; lighting, noise and sources of electrical interference; how conditions were kept consistent between sessions.]

4.6 Command Training and Calibration

[Aim: explain how participants prepared. Cover: which commands were trained; the training tool; number and length of training rounds; the criterion for being ready; how the same procedure was kept for everyone.]

4.7 Experimental Tasks and Protocol

[Aim: give the step-by-step procedure. Cover: each task, such as cued single commands and driving a set course; trial numbers and order; cue timing; rest breaks; rules for restarting or aborting. Put the full protocol in Appendix A.]

4.8 Pilot Study

[Aim: explain how the method was tested first. Cover: who took part; what you tried; what you changed as a result, such as thresholds, task length or instructions. Keep pilot data separate from the main results.]

4.9 Main Experimental Sessions

[Aim: describe how the main study ran. Cover: number of sessions and spacing between them; session structure and timing; who was present; how sessions were recorded; any deviations from the protocol.]

4.10 Data Collected and Availability

[Aim: state what data exists and who can access it. Cover: raw EEG if accessible, recognised commands, timestamps and the car's responses to tasks; file organisation and naming; pseudonymisation; secure storage and retention; what can be shared openly, with reproduction details in Appendix F. Check whether your Emotiv licence gives access to raw EEG.]

4.11 Evaluation Metrics

[Aim: define how performance is measured. Cover: how the metrics link to the hypotheses and objectives; shared definitions of trial, command window and correct response. For each metric below give a formula, the unit and the logged data used.]

4.11.1 Command Success Rate

[Aim: define this metric. Cover: correct commands executed as a share of commands cued; what counts as correct; the time window allowed.]

4.11.2 False Activation Rate

[Aim: define this metric. Cover: commands detected when none was intended, or a different one was; whether it is expressed per minute or as a share of trials, and why.]

4.11.3 Missed Command Rate

[Aim: define this metric. Cover: cued commands not detected within the time window, as a share of commands cued; how it relates to the success rate.]

4.11.4 Command Detection Latency

[Aim: define this metric. Cover: time from cue to detection by the framework; where each timestamp comes from; clock synchronisation; timing accuracy and how you verified it.]

4.11.5 End-to-End Response Time

[Aim: define this metric. Cover: time from cue to physical car response; how car movement is timestamped, such as onboard feedback or video; the breakdown into detection, processing and transmission delay.]

4.11.6 Task Completion Rate

[Aim: define this metric. Cover: tasks finished as a share of attempted tasks; the criteria for success and failure, such as time limits or leaving the course; any completion time measures.]

4.11.7 Performance Variation Across Sessions

[Aim: define how change is measured. Cover: the metrics compared across sessions; how learning, fatigue and variation between participants are examined; how you handle missing sessions.]

4.12 Data Analysis Methods

[Aim: explain the analysis before it is run. Cover: data cleaning and exclusion rules; descriptive statistics; statistical tests linked to each hypothesis, with assumption checks; significance level; effect sizes; software used. Choose tests suited to your sample size.]

4.13 Optional Investigation of a Custom Machine Learning Approach

[Aim: describe the optional comparison, if it is carried out. Cover: why a custom classifier could improve on built-in detection; available data; preprocessing and features; candidate models; validation that avoids leakage between sessions; comparison with Emotiv's detection. If it is not done, say so and remove Section 5.9.]

4.14 Validity, Reliability, and Reproducibility

[Aim: assess the method's strengths and threats. Cover: internal validity, such as learning and fatigue effects; external validity, such as sample and setting; reliability of measurement; how the open code, protocol and data allow reproduction.]

4.15 Chapter Summary

[Aim: close the chapter. Cover: the design, metrics and analysis in brief, leading into the results.]

5 Results

[Aim: open the chapter in one short paragraph. Cover: that it reports the findings without interpretation, which belongs in Chapter 6. Use only measured data.]

5.1 Introduction

[Aim: outline the chapter. Cover: how results are organised and how they follow the metrics in Section 4.11.]

5.2 Overview of Collected Experimental Data

[Aim: describe the data obtained. Cover: participants and sessions completed; total trials; exclusions and reasons; data quality issues; deviations from the protocol. A summary table works well.]

5.3 Performance of BCI Command Execution

[Aim: report how well commands were detected. Cover: overall and per-command results; a confusion matrix of cued versus detected commands; variation between participants.]

5.4 Success Rates and Missed Commands

[Aim: report these two rates. Cover: mean and spread for each command and participant; a table and a chart; the result of any hypothesis test on H1.]

5.5 False Activation Results

[Aim: report unintended commands. Cover: rates by participant and condition; which commands were most often triggered wrongly; any safety events or emergency stops.]

5.6 Detection Latency and End-to-End Response Time

[Aim: report the timing results. Cover: mean, median, spread and outliers; distribution plots; the breakdown of delay by stage; the result of any hypothesis test on H2.]

5.7 Results of Car Control Tasks

[Aim: report task performance. Cover: completion rates by task and participant; completion times; reasons for failure.]

5.8 Performance Variation across Sessions

[Aim: report changes over time. Cover: the metrics by session, shown in a line chart or table; participant differences; the result of any hypothesis test on H3.]

5.9 Results of a Custom Machine Learning Model

[Aim: report the optional comparison, if carried out. Cover: model performance with validation method; comparison with built-in detection using the same data; any offline-only caveat. Remove this section if the work was not done.]

5.10 Summary of Findings

[Aim: close the chapter with the main results. Cover: the key figures for each metric; the outcome for each hypothesis, as supported or not supported, without explaining why.]

6 Discussion

[Aim: open the chapter in one short paragraph. Cover: how it interprets the results against the research question.]

6.1 Introduction

[Aim: outline the chapter. Cover: restate the research question and the order of the sections that follow.]

6.2 Interpretation of Experimental Results

[Aim: say what the results mean. Cover: the main findings and their likely explanations; unexpected results; whether each hypothesis held and why. Distinguish what the data show from what you infer.]

6.3 Evaluation against Research Objectives

[Aim: assess the work against its objectives. Cover: for each objective in Section 1.7, the supporting evidence; a table may help. Answer the research question directly.]

6.4 Reliability of BCI-Based Car Control

[Aim: judge whether commands are dependable enough. Cover: what success, missed command and false activation rates mean for real use; which factors affected reliability, such as user, command or training.]

6.5 Latency and Suitability for Real-Time Control

[Aim: judge whether the system is fast enough. Cover: what the delays mean for driving a car; which stage contributes most delay; which parts could improve and which depend on Emotiv's detection.]

6.6 Trade-offs between Speed and Reliability

[Aim: discuss the central design tension. Cover: how stricter thresholds and smoothing reduce errors but add delay; what your data shows about this; the balance you recommend, and for what use.]

6.7 Implications for Consumer-Grade EEG Applications

[Aim: look beyond this car. Cover: what the results suggest for assistive technology, robotics and other uses of consumer EEG; where such devices are and are not appropriate, including safety-critical limits.]

6.8 Comparison with Previous Studies

[Aim: place results in the literature. Cover: comparison with the studies in Section 2.9, by device, paradigm and metric; agreements and differences; why direct comparison may be limited.]

6.9 Technical and Methodological Limitations

[Aim: assess the study honestly. Cover: sample size and participant mix; controlled environment; black-box detection; hardware and timing measurement limits; how each may have affected the results.]

6.10 Potential Improvements and Future Work

[Aim: suggest practical improvements to this system. Cover: changes to the framework, control logic, safety and protocol, each linked to a finding or limitation. Broader research directions belong in Section 7.6.]

6.11 Chapter Summary

[Aim: close the chapter. Cover: the main interpretation and its significance, leading into the conclusion.]

7 Conclusion and Future Work

[Aim: open the chapter in one short paragraph. Cover: that it brings the work together. Introduce no new results or sources.]

7.1 Introduction

[Aim: outline the chapter. Cover: restate the research question and the order of the sections that follow.]

7.2 Summary of the Research

[Aim: retell the work briefly. Cover: the problem; what was built; how it was tested; the main findings.]

7.3 Main Contributions

[Aim: state what the work delivered. Cover: the actual outcomes against the expected contributions in Section 1.10, including the framework, protocol, evidence and any data released.]

7.4 Extent to Which Objectives Were Achieved

[Aim: close the loop on the objectives. Cover: each objective from Section 1.7, marked as met, partly met or not met, with a short justification; a direct answer to the research question.]

7.5 Limitations of the Study

[Aim: state the limits on the conclusions. Cover: a brief summary of Section 6.9 and how far the findings can be generalised.]

7.6 Recommendations for Future Research

[Aim: suggest what other researchers should do next. Cover: larger and more diverse samples; custom classifiers; other paradigms or devices; less controlled environments; long-term studies. Base each on your findings.]

7.7 Final Remarks

[Aim: end the dissertation in one short paragraph. Cover: the wider significance of the work and the main message for the reader.]

References

Besl, P. and McKay, N. D. (1992). A method for registration of 3-d shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256.

Hassanien, A. E. and Azar, A. T., editors (2015). *Brain-Computer Interfaces: Current Trends and Applications*, volume 74 of *Intelligent Systems Reference Library*. Springer International Publishing, Cham.

Appendices

Appendix A — Experimental Protocol

[Aim: give the complete protocol. Include: step-by-step session script; participant instructions; task definitions and cue timing; session checklist; abort and emergency procedures.]

Appendix B — Participant Information and Informed Consent Documents

[Include only if applicable. Aim: provide the ethics documents. Include: participant information sheet; blank consent form; questionnaires; the ethics approval reference, if granted. Exclude anything that identifies a participant.]

Appendix C — Supplementary Experimental Results

[Aim: hold detail too long for Chapter 5. Include: per-participant and per-session tables; extra charts; full statistical outputs.]

Appendix D — System Architecture and Technical Configuration

[Aim: give the full technical record. Include: detailed diagrams; car wiring or hardware details; configuration parameters and thresholds; message format; software versions.]

Appendix E — Source Code and Repository Information

[Aim: point readers to the code. Include: repository URL; the release or commit used in the study; licence; folder structure; any key listings. Never include API credentials.]

Appendix F — Instructions for Replicating the Experimental Data

[Aim: let others repeat the analysis. Include: where the anonymised data is stored and how to access it; data dictionary; installation steps; commands that reproduce each result.]

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